Quantum Multiple Rotation Averaging

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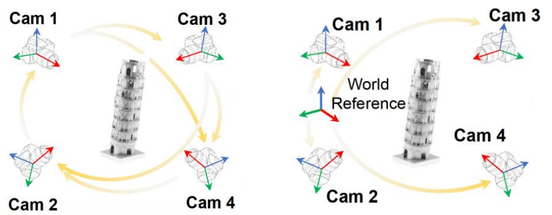
3DV 2026

Multiple Rotation Averaging (MRA)



$$\min_{R_1,\dots,R_N \in \mathrm{SO}(3)} \sum_{(i,j)} \|\tilde{R}_{ij} R_i - R_j\|_F^2$$

Motivation

- Parallel optimization landscape exploration enabled by quantum superposition;
- Direct hardware execution of problems at nano-milli (seconds) timescales per sample;
- Investigate potentials of quantum computing in vision (or related) problems.

MRA on Quantum Annealers (Scheme)

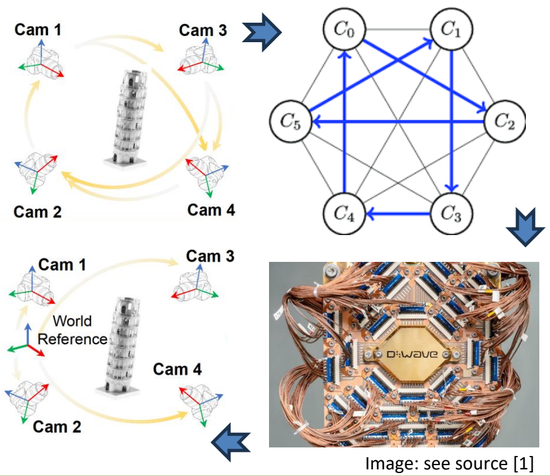


Image: see source [1]

Reference

[1] https://www.it-daily.net/shortnews/d-wave-quantencomputer

Background/ Quantum Multiple Rotation Averaging (QMRA) Theories

Background on Optimization with Quantum Annealers

Combinatorial Quadratic Unconstrained Binary Optimization (QUBO) Problems:

$$\min_{x\in\{0,1\}^n} f(x) \qquad f(x) = x^\top Q x + c^\top x,$$

Physically, this maps to an Ising Hamiltonian whose ground state encodes the optimal soln:

$$H_P = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j,$$

Quantum annealers solve this through adiabatic evolution via sampling:

$$H(t) = (1 - s(t))\, H_0 + s(t) H_P, \quad s(t) \in [0,1],$$

Quadratic Optimization Problem

Quadratic optimization problem (QOP):

$$\boxed{\min_{R_1,\dots,R_N} \sum_{(i,j)} \|\tilde{R}_{ij} R_i - R_j\|_F^2 = \min_R R^\top Q R}$$

with $R = [\cdots \mathrm{vec}(R_i)^\top \cdots]^\top \in \mathbb{R}^{9N\times 1}$

and $Q = -2 \begin{bmatrix} I\otimes R_{11}^\top & \cdots & I\otimes R_{1N}^\top \\ \vdots & \ddots & \vdots \\ I\otimes R_{N1}^\top & \cdots & I\otimes R_{NN}^\top \end{bmatrix}$

Lie Algebra: SO(3) – so(3) Representation

Representation of rotation as an exponential map:

$$R_i(\boldsymbol{v}_i) = \exp(\mathcal{R}(\boldsymbol{v}_i)) = I + \frac{\sin\theta}{\theta}\mathcal{R}(\boldsymbol{v}_i) + \frac{1-\cos\theta}{\theta^2}\mathcal{R}^2(\boldsymbol{v}_i)$$

tangent vector of a rotation $\boldsymbol{v}_i = (v_i^1, v_i^2, v_i^3)^\top \in \mathbb{R}^3$

$\mathcal{R} : \mathbb{R}^3 \to \mathfrak{so}(3)$ is Lie algebra isomorphism

$$\mathcal{R}(\boldsymbol{v}_i) := \begin{pmatrix} 0 & -v_i^3 & v_i^2 \\ v_i^3 & 0 & -v_i^1 \\ -v_i^2 & v_i^1 & 0 \end{pmatrix}, \quad \mathcal{R}(\boldsymbol{v}_i)^\top = -\mathcal{R}(\boldsymbol{v}_i)$$

Linearization/ Iterative Solver

Iteratively optimise over the difference:

$$\Delta\boldsymbol{v}^{k+1} = \arg\min_{\Delta\boldsymbol{v}\in\mathbb{R}^{3N}} \hat{R}^k(\Delta\boldsymbol{v})^\top Q \hat{R}^k(\Delta\boldsymbol{v}),$$

$$\boldsymbol{v}^{k+1} = \boldsymbol{v}^k + \Delta\boldsymbol{v}^{k+1}, \quad \textit{preserved Lie group}$$

the next guess *the current guess* *difference*

with the first-order Taylor linearisation (due to non-linearity of the exponential map):

$$\hat{R}^k(\Delta\boldsymbol{v}) := R(\boldsymbol{v}^k) + \nabla R(\boldsymbol{v}^k)^\top \Delta\boldsymbol{v}$$

Bounded Search with Binary Encoding

To satisfy the infinity norm constraint

$$\|\Delta\boldsymbol{v}\|_\infty \le \delta^k \qquad \Delta\boldsymbol{v} = -\delta^k \mathbf{1}_{3N} + D\boldsymbol{q},$$

discretise $[-\delta^k, \delta^k]$ uniformly using m qubits per dimension:

$$(\Delta\boldsymbol{v})_i = -\delta^k + \frac{2\delta^k}{s}\sum_{\ell=0}^{m-1} 2^\ell q_{i,\ell}$$

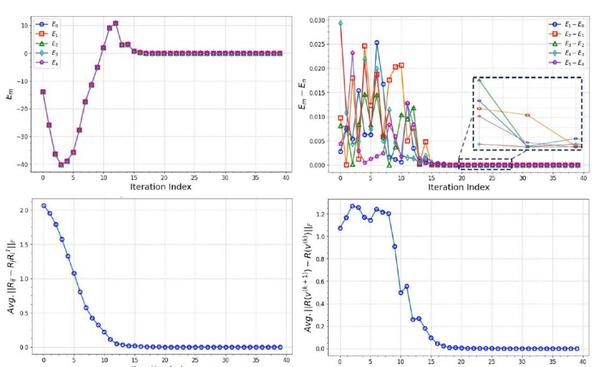
where $s = 2^m - 1$.

Finally, we arrive at

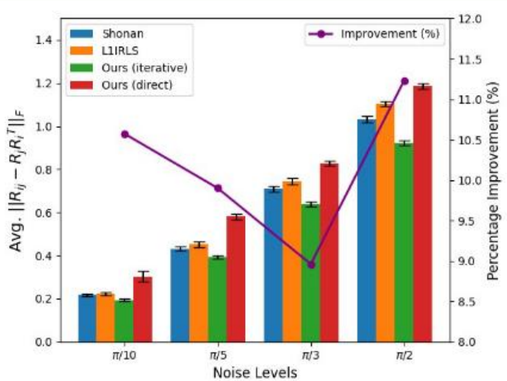
$$\min_{R_1,\dots,R_N} \sum_{(i,j)} \|\tilde{R}_{ij} R_i - R_j\|_F^2 = \arg\min_{\boldsymbol{q}\in\mathbb{B}^{3Nm}} \boldsymbol{q}^\top \tilde{Q} \boldsymbol{q} + \boldsymbol{q}^\top \tilde{\boldsymbol{c}},$$

Results / Supported Applications

Visualization of the Iterative Process



Tabulated Results



Structure-from-Motion Use-case

For more details, please check our website: https://4dqv.mpi-inf.mpg.de/QMRA/.

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